

THABOLEZWE MABANDLA

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SUMMARY

A graduate student in Engineering Artificial Intelligence at Carnegie Mellon University Africa, with strong software engineering fundamentals and hands-on experience in Python, distributed systems, and ML infrastructure. Built production-grade data pipelines and fine-tuned large language models for applied use cases, including real-time analytics and threat-intelligence systems. Currently using graduate studies to grow expertise in Applied AI, Embedded Systems and Robotics.

EDUCATION

Masters of Science in Engineering AI August 2025 - May 2027
Carnegie Mellon University, Africa
Concentration: Applied AI

BACHELOR OF SCIENCE HONOURS DEGREE IN AI & MACHINE LEARNING August 2020 - June 2024
University of Zimbabwe, Harare
Concentration: Natural Language Processing
Thesis Title: Digitization of Indigenous Zimbabwean Languages: A Shona Spell Checker
Graduated with First Class Honors
Awarded the University of Zimbabwe Book Prize
Awarded the Best Undergraduate Researcher and Innovator in the Faculty of Computer Engineering

EXPERIENCE

Backend Developer/ Machine Learning Engineer March 2025 - August 2025
CyberAlliance, Raleigh, North Carolina

- Contributed to SallyAI, a threat intelligence platform delivering real-time cybersecurity insights via an LLM agent and code agent.
- Engineered scalable data pipelines with Apache Airflow, Kafka, Redis, and Postgres, improving workflow automation and reliability on distributed systems.
- Fine-tuned LLMs (HuggingFace Transformers, BitsandBytes) using MLFlow and AWS, optimizing inference efficiency and reducing training costs.
- Implemented RAG pipelines with Weaviate, Elastic Stack, and Sentence Transformers, enhancing retrieval accuracy and contextual search for enterprise clients.
- Deployed a MISP threat intelligence server for real-time ingestion and sharing of cyber threat data across teams.
- Integrated multiple CTI clients (NVD NIST, Abuse.ch, AlienVault OTX, MITRE.org), streamlining security workflows and strengthening proactive defense capabilities.

AI Software Developer October 2024 - March 2025
Rubiem Innovations, Harare, Harare

- Led the development of a Local LLM agent prototype for Zimpapers (Zimbabwe's largest media house), enhancing journalists' article-writing workflow with enhanced retrieval from internal knowledge bases and integrated Google Search capabilities.
- Designed and deployed LLM-based applications that automate customer support and content generation for enterprise clients.
- Assisted in the process of providing AI strategy guidance for banks and telecoms, aligning organizational needs with scalable AI adoption.
- Translated technical insights into actionable recommendations for non-technical stakeholders, while delivering training and ongoing client support.
- Conducted client visits to gather business requirements, propose tailored AI solutions, and ensure alignment between technical deliverables and client needs.

Research Scientist

January 2024 - June 2024

Zimbabwe Centre for High Performance Computing, Harare

- Conducted NLP research for developing language technologies for Zimbabwe's two main indigenous languages.
- Developed and deployed a novel Shona spelling correction LLM, the first of its kind for one of Zimbabwe's main native languages.
- Built machine learning workflows on HPC clusters, accelerating large-scale model training and experimentation.

IT Operations Intern

September 2022 - August 2023

Old Mutual & CABS, Harare

- Monitored and optimized server uptime, network connectivity, and application performance, ensuring continuous availability of financial systems.
- Partnered with software teams to deploy and update applications, reducing deployment errors and improving turnaround time.
- Authored and maintained technical documentation, standardizing IT operations across departments.
- Executed routine system maintenance and security compliance checks, strengthening infrastructure reliability and regulatory adherence.

PROJECTS

LLM for Shona Spelling Correction - Undergrad Research

August 2023 - June 2024

- Developed the first LLM-based spell checker for Shona, one of Zimbabwe's major native languages.
- Built and fine-tuned models on HPC clusters, creating tools that support language digitization and preservation.
- Contributed to ongoing research on low-resource NLP and African language technology.

LLM Agent Prototype for Zimpapers

October 2024 - December 2024

- Designed a local LLM-powered agent to assist journalists by retrieving from internal archives and integrating real-time Google Search.
- Improved the article writing process by enabling contextual knowledge access and faster research workflows.
- Presented prototype to executives at Zimbabwe's largest media house, demonstrating business impact.

IoT + AI for Power Infrastructure Security

July 2024 - July 2024

- Collaborated as part of Team BitLords, winning the 2024 TEXPO Hackathon competition out of 8 teams.
- Designed an IoT-based system to detect and signal transformer oil and copper cable theft in real time for ZESA (Zimbabwe Electricity Supply Authority).
- Collected and analyzed multi-location sensor data to train AI models for predicting theft incidents before they occur.
- Delivered a prototype that provided actionable insights to ZESA through sensor analytics and predictive modeling.

SKILLS

- Programming & Scripting:** (advanced) Python, (basic) JavaScript, SQL, Bash.
- Machine Learning & AI:** HuggingFace Transformers, PyTorch, TensorFlow, Scikit-learn, NLP, LLMOps.
- Data Engineering & Databases:** Apache Airflow, Kafka, Redis, Postgres, Elastic Stack, Weaviate, Docker, Ubuntu Server, AWS.
- Tools & Platforms:** Git, MLFlow, BitsandBytes, Google Cloud, Canva.

INTERESTS

World Models, Computer Vision, Vision Language Models, Drone Technologies, Robotics, Language Technologies